

High-Dimensional Causal Inference

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Outline

- 1 Preliminaries
- 2 High-Dimensional AIPW
- 3 Debiased IPW [WS24]
- 4 High-Dimensional Discrete Covariates [Zen+24]

Caveats

- We will focus exclusively on counterfactual/potential outcomes framework
- This excludes
 - Graphical Methods
 - Targeted MLE
 - <https://www.khstats.com/blog/tmle/tutorial>
 - <https://tlverse.org/tlverse-handbook/tmle3.html>

Papers of Focus

- *“Debiased Inverse Propensity Score Weighting for Estimation of Average Treatment Effects with High-Dimensional Confounders”*
 - Yuhao Wang & Rajen Shah [WS24]
- *“Causal Inference with High-dimensional Discrete Covariates”*
 - Zhenghao Zeng, Sivaraman Balakrishnan, Yanjun Han, Edward H. Kennedy [Zen+24]

Typical Causal Set-Up

- Estimand is ATE, $\tau = \mathbb{E}(Y(1) - Y(0))$
- Observe $T \in \{0, 1\}^N$, $\mathbf{X} \in R^{N \times d}$ pre-treatment covariates
- Common assumptions:

Unconfoundedness: $\{Y(1), Y(0)\} \perp\!\!\!\perp T \mid \mathbf{X}$

SUTVA: $Y_i = Y_i(1)Z_i + Y_i(0)(1 - Z_i)$

Positivity: $\mathbb{P}(T_i = 1 \mid \mathbf{X}_i) =: \pi(\mathbf{x}) \in [\epsilon, 1 - \epsilon]$ for $f(\mathbf{x}) > 0$

AIPW

- IPW estimator $\hat{\tau}_{\text{IPW}} := \frac{1}{n} \sum_{i=1}^n \frac{T_i Y_i}{\hat{\pi}(X_i)} - \frac{1}{n} \sum_{i=1}^n \frac{(1-T_i) Y_i}{1-\hat{\pi}(X_i)}$
 - If $\hat{\pi} \xrightarrow{P} \pi$ consistent
- AIPW estimator $\hat{\tau}_{\text{AIPW}} = \frac{1}{n} \sum_{i=1}^n \frac{T_i(Y_i - \mu_i)}{\hat{\pi}(X_i)} - \frac{1}{n} \sum_{i=1}^n \frac{(1-T_i)(Y_i - \mu_i)}{1-\hat{\pi}(X_i)}$
 - μ is some/any “augmentation” that ideally retains unbiasedness and reduces variance of our estimator
 - $\mu \perp T \mid X$ retains unbiasedness
 - $\hat{\mu} = (1 - \hat{\pi}(X))\hat{r}_1(X) + \hat{\pi}(X)\hat{r}_0(X)$ is the common “AIPW”
 - For $r_j(X) = \mathbb{E}(Y(j) \mid X) = \mathbb{E}(Y \mid T = j, X)$

AIPW cont'd

- One can show $\sqrt{n}(\tau_{\text{AIPW}} - \tau) \xrightarrow{d} N(0, V)$
- Comparing $\hat{\tau}_{\text{AIPW}}$ to an oracle (in π, μ) τ_{AIPW} , we have

$$\begin{aligned} & |\hat{\tau}_{\text{AIPW}} - \hat{\tau}_{\text{AIPW}}^*| \\ &= O_P \left(\max_{w \in \{0,1\}} \mathbb{E} \left[(\hat{r}_w(X_i) - r_w(X_i))^2 \right]^{\frac{1}{2}} \mathbb{E} \left[(\hat{e}(X_i) - e(X_i))^2 \right]^{\frac{1}{2}} \right) \end{aligned}$$

- If r_j, π are $n^{-1/4}$ estimable, $\sqrt{n}(\hat{\tau}_{\text{AIPW}} - \tau) \xrightarrow{d} N(0, V)$ and $\hat{\tau}_{\text{AIPW}}$ is semiparametrically efficient

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“Double-Selection” Methods

When do we remain doubly-robust while performing model selection?

- “Double-selection” methods - Lasso variable selection followed by unpenalized but supplemented re-fitting
 - Under a (partially-) linear model $Y = \tau T + f(X) + \epsilon$, can recover asymptotic normality under double-Lasso selection [BCH12]
 - Requires $s_\pi \vee s_r = o(\sqrt{n}/\log p)$
 - See pg. 11 for details on implementation
<https://arxiv.org/pdf/1201.0224>
 - Similar work by [Far15] with stricter $s_\pi s_r = o(\sqrt{n}/\log(p)^{1.5+\delta})$, $\delta > 0$ using group lasso, double-selection style
 - See pg. 21 for procedure, pg. 7 Corollary 1 for sparsity requirements
<https://arxiv.org/pdf/1309.4686>

High-level, these methods require both r, π to be $\sqrt{n}/\log(p)$ -sparse

Exploiting Sparsity Structure or “De-biasing” Methods

- [BWZ19] assume linear-logistic model and ultra-sparsity if *either* model
 - $\hat{\theta}_{(w)} = \operatorname{argmin}_{\theta} \left\{ \frac{1}{n} \sum_{i=1}^n (\mathbb{1}\{W_i \neq w\} X_i' \theta + \mathbb{1}\{W_i = w\} \exp(-X_i' \theta)) + \lambda_{\theta} \|\theta\|_1 \right\}$
 - $\hat{\beta}_{(w)} = \operatorname{argmin}_{\beta} \left\{ \frac{1}{n} \sum_{W_i=w} \exp(-X_i' \hat{\theta}_{(w)}) (Y_i - X_i' \beta)^2 + \lambda_{\beta} \|\beta\|_1 \right\}$
 - $\tau = \tau_1 - \tau_0$ for

$$\hat{\tau}_1 = \frac{1}{n} \sum_{i=1}^n \left[\left(X_i' \hat{\beta}_{(1)} + W_i \left[1 + \exp(-X_i' \hat{\theta}_{(1)}) \right] \right) (Y_i - X_i' \hat{\beta}_{(1)}) \right]$$
 - Bias can decompose as $|\hat{\tau}_1 - \tau| \leq \left\| n^{-1} \sum_{i=1}^n \left[1 - W_i \left(1 + \exp(-X_i' \hat{\theta}_{(1)}) \right) \right] X_i \right\|_{\infty} \left\| \hat{\beta}_{(1)} - \beta_{(1)} \right\|_1$

Exploiting Sparsity Structure or “De-biasing” Methods

- [AIW18]¹ require only $s_r = o(\sqrt{n}/\log(p))$ Athey paper
<https://arxiv.org/pdf/1604.07125>
 - Estimate outcome coefficients $\hat{\beta}$ by lasso
 - Estimate balancing weights γ (see pg. 6/7)
 - $\hat{\mu}_c = \bar{X}_t \cdot \hat{\beta}_c + \sum_{\{i: W_i=0\}} \gamma_i \left(Y_i^{\text{obs}} - X_i \cdot \hat{\beta}_c \right)$
 - $|\hat{\mu}_c - \mu_c| \leq \|\bar{X}_t - \mathbf{X}_c^\top \gamma\|_\infty \|\hat{\beta}_c - \beta_c\|_1 + \left| \sum_{\{i: W_i=0\}} \gamma_i \varepsilon_i \right|$
 - Here $\eta = Y(0) - X^\top \beta_c$, i.e. outcome regression noise
 - Resulting estimator is $\sqrt{n}$ -consistent and asymptotically normal under additional technical conditions

¹Notation here for their primary endpoint, ATT, but extendable to ATE

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Estimator

Estimand $\tau = \mathbb{E}(Y(1) - Y(0))$

$$\hat{\tau}_{\text{DIPW}} := \frac{1}{n} \sum_{i=1}^n \left(\frac{T_i (Y_i - \hat{\mu}_i)}{\hat{\pi}_i} - \frac{(1 - T_i) (Y_i - \hat{\mu}_i)}{1 - \hat{\pi}_i} \right)$$

- $\hat{\pi}$ evaluated via lasso logistic regression
- $\hat{\mu}$ is evaluated via a quadratic program, an “orthogonalization” of the AIPW style augmentation
- Requires only $s_{\pi} = o(\sqrt{n}/\log p)$ for consistency, $o(1/\sqrt{\log n})$ estimation of regression models achieves semiparametric efficiency
 - Equivalent to $s_r = o(n/[\log n \log p])$ requirement
 - **Caveat:** Inference (i.e. CI's) require s_{π} assumptions

Estimation Procedure Outline

- Estimate $\hat{\pi} = \psi(x^T \hat{\gamma})$ with $\hat{\gamma}$ estimated via lasso on auxiliary data $\mathcal{D}_B$
 - Assumption: $s_{\pi} = o(\sqrt{n} \log(p))$
- Construct $\tilde{\mu}$, an estimate of μ_{ORA} , using $\mathcal{D}_B$
- Construct $\hat{\mu}$ using $\mathcal{D}_A$ by convex program above
- Plug-in and estimate $\hat{\tau}_{\text{DIPW}}$, AIPW style estimator with $\hat{\mu}, \hat{\pi}$

Estimation Procedure

- Observe n , iid $(X, Y, T) \in \mathbb{R}^p \times \mathbb{R} \times \{0, 1\}$, say $\mathcal{D} = (\mathbf{X}, \mathbf{Y}, \mathbf{T})$
 - We will also use auxiliary datasets $\mathcal{D}_A = (\mathbf{X}_A, \mathbf{Y}_A, \mathbf{T}_A), \mathcal{D}_B$
 - Assume X, Y are σ_Y^2, σ_X^2 sub-Gaussian and $\max_{t \in \{0, 1\}} |\mathbb{E}Y(t)| < m_Y$
- Assume a logistic model for the propensity
$$\pi(x) = \mathbb{P}(T = 1 \mid X = x) = \psi(x^T \gamma) := (1 + \exp(-x^T \gamma))^{-1}$$
- Let $\mu_{\text{ORA}}(x) := (1 - \pi(x))r_1(x) + \pi(x)r_0(x)$
 - Recall $\mathbb{E}\tau_{\text{ORA}} = \tau$ if $\mu \perp T \mid X$

Estimation Procedure

- Let $\tilde{Y}_i := \frac{T_i Y_i (1 - \hat{\pi}_i)}{\hat{\pi}_i} + \frac{(1 - T_i) Y_i \hat{\pi}_i}{1 - \hat{\pi}_i}$
- Estimate $\hat{\gamma}$ using $\mathcal{D}_B$
- Bias in $\hat{\tau}_{IPW}$ then becomes² determined by

$$\approx \left| \frac{1}{n} \sum_{i=1}^n (\tilde{Y}_i - \mu_i) \mathbf{X}_i^\top (\hat{\gamma} - \gamma) \right| \leq \frac{1}{n} \left\| \mathbf{X}^\top \tilde{\mathbf{Y}} - \mathbf{X}^\top \boldsymbol{\mu} \right\|_\infty \|\hat{\gamma} - \gamma\|_1$$

- $\|\hat{\gamma} - \gamma\|_1 = o(s\sqrt{\log p/n})$ whp
- So to control bias of $\hat{\tau}_{DIPW}$, we need only study $\frac{1}{n} \left\| \mathbf{X}^\top \tilde{\mathbf{Y}} - \mathbf{X}^\top \boldsymbol{\mu} \right\|_\infty$

²From comparing $\hat{\tau}_{IPW} - \tau_{ORA}$

Estimation Procedure

$$\begin{aligned} \frac{1}{n} \left\| \mathbf{X}^\top \tilde{\mathbf{Y}} - \mathbf{X}^\top \boldsymbol{\mu} \right\|_\infty &\leq \left\| \frac{1}{n_A} \mathbf{X}_A^\top \left\{ \tilde{\mathbf{Y}}_A - f(\mathbf{X}_A) \right\} - \frac{1}{n} \mathbf{X}^\top \left\{ \boldsymbol{\mu} - f(\mathbf{X}) \right\} \right\|_\infty \\ &\quad + \left\| \frac{1}{n_A} \mathbf{X}_A^\top \left\{ \tilde{\mathbf{Y}}_A - f(\mathbf{X}_A) \right\} - \frac{1}{n} \mathbf{X}^\top \left\{ \tilde{\mathbf{Y}} - f(\mathbf{X}) \right\} \right\|_\infty \end{aligned}$$

- First term allows us to approximate $\boldsymbol{\mu}$ by $\mathcal{D}_A$, respecting the requirement $\boldsymbol{\mu} \perp T \mid X$
- Second term $\leq c \sqrt{\log p / \min\{n, n_A\}}$ under sub-Gaussian assumptions on X, Y, X_A, Y_A
- f is a fixed function, which we will choose

Estimation Procedure

- Considering $\left\| \frac{1}{n_A} \mathbf{X}_A^\top \left\{ \tilde{\mathbf{Y}}_A - f(\mathbf{X}_A) \right\} - \frac{1}{n} \mathbf{X}^\top \{ \boldsymbol{\mu} - f(\mathbf{X}) \} \right\|_\infty \leq \eta$
- Select $\eta \asymp \sqrt{\log(p)/n}$, then

$$\| \mathbf{X}^T \tilde{\mathbf{Y}} - \mathbf{X}^T \boldsymbol{\mu} \|_\infty \| \hat{\gamma} - \gamma \|_1 \leq \sqrt{\log(p)/n} \cdot s \sqrt{\log(p)/n} = s \log(p)/n$$

- $o(n^{-1/2})$ under $x = o(\sqrt{n}/\log(p))$
- Remains to identify f, μ

Estimation Procedure

- **Lemma 1:** μ_{ORA} minimizes $V(\tau_{\text{ORA}})$.
 - So ideally $\mu \approx \mu_{\text{ORA}}$, but cannot regress $\tilde{\mathbf{Y}} \sim \mathbf{X}$ (as we require $\mu \perp \mathbf{T} \mid \mathbf{X}$)
 - Construct $\tilde{\mu}$ using $\mathcal{D}_B$, then estimate $\mu = \operatorname{argmin} \|\mu - \tilde{\mu}(\mathbf{X})\|_2^2$, using $\tilde{\mu} = f$

Thus estimate $\hat{\mu}$ by the convex program

$$\begin{aligned} \hat{\mu} &= \operatorname{argmin}_{\mu \in \mathbb{R}^n} \frac{1}{n} \|\tilde{\mu}(\mathbf{X}) - \mu\|_2^2 \\ \text{subject to } &\left\| \frac{1}{n_A} \mathbf{X}_A^\top \left\{ \tilde{\mathbf{Y}}_A - \tilde{\mu}(\mathbf{X}_A) \right\} - \frac{1}{n} \mathbf{X}^\top \{ \mu - \tilde{\mu}(\mathbf{X}) \} \right\|_\infty \leq \eta \end{aligned}$$

Estimation Procedure Outline

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 - Assumption: $s_{\pi} = o(\sqrt{n} \log(p))$
- Construct $\tilde{\mu}$, an estimate of μ_{ORA} , using $\mathcal{D}_B$
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Inference

- See Theorem 3 (pg. 11) for asymptotic normality result and subsequent CI construction
 - Note new dependence on $\|\mu - \mu_{\text{ORA}}\|_{\infty}$ (both) and $s_{\pi} = o(\sqrt{n}/\log(p))$ assumption (CI construction only)

Additional Notes

- Asymptotic near-normality result $\sqrt{n}(\hat{\tau}_{DIPW} - \tau) = \delta + \sigma_{\mu}\zeta_1 + \sigma\zeta_2$ conditional on $\mathcal{D}$
 - $\delta < c(s + \sqrt{s \log(n)} \log(p)/\sqrt{n})$ whp
 - η is “near-Normal” in a Berry-Esseen sense, see Theorems 2 & 3
- Can use a sample-splitting procedure in place of hold-out/auxiliary data sets
- Can extend to link functions (say ϕ) beyond $\psi(x) = (1 + \exp(-x))^{-1}$, with conditions on ϕ', ϕ''

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Set-Up

- Observe iid $(Y, X, A) \in \{0, 1\} \times \{0, \dots, d\}^K \times \{0, 1\}$
 - $P(X = k) = p_k, k \in [d]$
 - $A \mid X = k \sim \text{Ber}(\pi_k)$
 - $Y \mid X = k, A = a \sim \text{Ber}(\mu_{ak})$
 - $q_{ak} = \mathbb{P}(X = k, A = a, Y = 1) = p_k [a\pi_k + (1 - a)(1 - \pi_k)] \mu_{ak}$
 - $w_k = \mathbb{P}(X = k, A = 1) = p_k \pi_k$
- Estimand is typical
$$\psi = \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[\mathbb{E}[Y|X, A = 1] - \mathbb{E}[Y|X, A = 0]]$$
- Typical causal assumption, here positivity is on $\pi_k, k \in [d]$
 - Interesting/specific consideration of ϵ here as we might expect $\epsilon_n \rightarrow 0$ as $n \rightarrow \infty$ see Remark 1

Estimator Equivalence

Consider our suite of classical estimators

$$\hat{\psi} = \sum_{k=1}^d \hat{p}_k (\hat{\mu}_{1k} - \hat{\mu}_{0k}) = \sum_{k=1}^d \hat{p}_k \left(\frac{\hat{q}_{1k}}{\hat{w}_k} - \frac{\hat{q}_{0k}}{\hat{p}_k - \hat{w}_k} \right)$$

$$\hat{\psi}_{\text{reg}} = \mathbb{P}_n [\hat{\mu}_{1X} - \hat{\mu}_{0X}],$$

$$\hat{\psi}_{\text{ipw}} = \mathbb{P}_n \left[\frac{AY}{\hat{\pi}_X} - \frac{(1-A)Y}{1 - \hat{\pi}_X} \right],$$

$$\hat{\psi}_{\text{dr}} = \mathbb{P}_n \left[\frac{A(Y - \hat{\mu}_{1X})}{\hat{\pi}_X} + \hat{\mu}_{1X} - \frac{(1-A)(Y - \hat{\mu}_{0X})}{1 - \hat{\pi}_X} - \hat{\mu}_{0X} \right]$$

Claim: $\hat{\psi} = \hat{\psi}_{\text{Reg}} = \hat{\psi}_{\text{IPW}} = \hat{\psi}_{\text{AIPW}}$ for $\hat{\psi}$ constructed using sample-average plug-in estimators for μ, π, q, w

So we consider only

$$\begin{aligned}\hat{\psi} &= \sum_{k=1}^d \hat{p}_k (\hat{\mu}_{1k} - \hat{\mu}_{0k}) = \sum_{k=1}^d \hat{p}_k \left(\frac{\hat{q}_{1k}}{\hat{w}_k} - \frac{\hat{q}_{0k}}{\hat{p}_k - \hat{w}_k} \right) \\ &= \hat{\psi}_1 - \hat{\psi}_0 \\ \psi &= \sum_{k=1}^d p_k (\mu_{1k} - \mu_{0k}) = \sum_{k=1}^d p_k \left(\frac{q_{1k}}{w_k} - \frac{q_{0k}}{p_k - w_k} \right) \\ &= \psi_1 - \psi_0\end{aligned}$$

Estimation Rates

- See Proposition 3 for bias-derivation, requires $d = o(n)$ scaling
- See Proposition 4 for minimax lower bound contains $\frac{d^2}{n^2 \log^2 n}$ terms, that is $\hat{\psi}$ is minimax optimal up to log factors

References I

- [AIW18] Susan Athey, Guido W. Imbens, and Stefan Wager. *Approximate Residual Balancing: De-Biased Inference of Average Treatment Effects in High Dimensions*. Jan. 31, 2018. [arXiv: 1604.07125\[econ,math,stat\]](#).
- [BCH12] Alexandre Belloni, Victor Chernozhukov, and Christian Hansen. *Inference on Treatment Effects After Selection Amongst High-Dimensional Controls*. May 9, 2012. [arXiv: 1201.0224\[econ,stat\]](#).
- [BWZ19] Jelena Bradic, Stefan Wager, and Yinchu Zhu. *Sparsity Double Robust Inference of Average Treatment Effects*. May 2, 2019. [arXiv: 1905.00744\[econ,math,stat\]](#).

References II

- [Far15] Max H. Farrell. “Robust Inference on Average Treatment Effects with Possibly More Covariates than Observations”. In: *Journal of Econometrics* 189.1 (Nov. 2015), pp. 1–23. arXiv: 1309.4686[econ,math,stat].
- [WS24] Yuhao Wang and Rajen D. Shah. *Debiased Inverse Propensity Score Weighting for Estimation of Average Treatment Effects with High-Dimensional Confounders*. Apr. 11, 2024. arXiv: 2011.08661[math,stat].
- [Zen+24] Zhenghao Zeng et al. *Causal Inference with High-dimensional Discrete Covariates*. May 5, 2024. arXiv: 2405.00118[math,stat].

Misc. Undiscussed Papers I

- “Debiasing the Lasso: Optimal Sample Size for Gaussian Designs” - Adel Javanmard, Andrea Montanari
<https://arxiv.org/pdf/1508.02757>
 - Discusses gap between lasso rate $s = o(n/\log p)$ and $s = o(\sqrt{n}/\log p)$ requirement in Slide 9 refs
- “Deep Neural Networks for Estimation and Inference” - Max H. Farrell, Tengyuan Liang, Sanjog Misra
<https://arxiv.org/pdf/1809.09953>
 - Establishes conditions for semi-parametric efficiency of ATE when using deep NN's
- “Program Evaluation and Causal Inference with High-Dimensional Data” - Belloni, Chernozhukov, Fernandez-Val, Hansen
<https://arxiv.org/pdf/1311.2645>
 - Contemporary with some of the “double-selection” methods discussed